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Phlebotomine sand flies (Diptera: Psychodidae) of Morocco: results of an entomological survey along three transects from northern to southern country

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PEER REVIEW

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Comments

In this work authors conducted an entomological survey along three transects of Morocco where foci of cutaneous leishmaniasis occur. The data obtained gives practical information and suggests the need for a continuous entomological surveillance in this country.

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ABSTRACT

Objective: To study the sandflies distribution within their climatic and ecological context in three transects of Morocco: Ouarzazat-Mhamid, Fom Zguid-Marrakesh and Erfoud-Nador.

Methods: In total, twenty-nine stations were prospected, through 1800 Km, including four zoonotic cutaneous leishmaniasis foci (Ouarzazat, Zagora, Tata and Errachidia). Sand flies were collected using sticky paper traps for one night.

Results: Overall, 7140 sandflies were collected along the three transects. In the combined collections, nine *Phlebotomus* species: *Phlebotomus papatasi* (27.6%), *Phlebotomus longicuspis* (19%), *Phlebotomus sergenti* (18.2%), *Phlebotomus perniciosus* (6.2%), *Phlebotomus bergeroti* (2.9%), *Phlebotomus alexandri* (1.4%), *Phlebotomus chadlii* (0.8%), *Phlebotomus chabaudi* (0.5%) and *Phlebotomus ariasi* (0.5%) and five *Sergentomyia* species: *Sergentomyia minuta* (10.4%), *Sergentomyia fallax* (8.1%), *Sergentomyia dreyfussi* (2.1%), *Sergentomyia christophersi* (1.7%) and *Sergentomyia africana* (0.5%) were detected.

Conclusions: We update the entomological data in zoonotic cutaneous leishmaniasis foci and discuss the possible effect of many ecological factors as bioclimate, biotopes and altitude on the diversity and distribution of caught species.

KEYWORDS

Sandflies, Entomological survey, Ecology, Bioclimate, Morocco

1. Introduction

Phlebotomine sandflies (Diptera: Psychodidae) are the vectors

for leishmaniasis as well as for arboviruses and bartonellosis and responsible of sandfly fever, summer meningitis, vesicular stomatitis, Chandipura virus encephalitis and Carrion's disease[1].

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In Morocco, leishmaniasis are endemic diseases posing a major threat to public health. In 2011, Moroccan Ministry of Health reported 4 319 cases of human cutaneous leishmaniasis (CL) and 107 cases of visceral leishmaniasis (VL)[2]. CL caused by *Leishmania major* (*L. major*)(Kinetoplastida: Trypanosomatidae) is the most dominant form in the country with more than 24 450 cases reported in the last decade[2].

Phlebotomus papatasi (*P. papatasi*) and *Phlebotomus sergenti* (*P. sergenti*) are known to be the common vectors spreading *L. major* and *Leishmania tropica* (*L. tropica*), respectively in Morocco.

In the Mediterranean countries, *Leishmania infantum* (*L. infantum*) is the etiologic agent of VL[3]. While the subgenus “*Larrousius*” species: *Phlebotomus perniciosus* (*P. perniciosus*), *Phlebotomus ariasi* (*P. ariasi*) and *Phlebotomus longicuspis* (*P. longicuspis*) are considered as *L. infantum* vectors[4].

Moreover, Asmae et al. (2014) have demonstrated the coexistence of *L. tropica* and *L. infantum* as causative agents of CL in the Sefrou province, northeast of the country[5]. Knowledge of the distribution of vectors is important in predicting the spatial variations in the risk of disease. Previous studies in Morocco[6-10] showed that the distribution of sandflies was due to the bioclimate in great part. Current findings in Morocco showed that altitude (through the gradient on temperature, pressure and precipitation) and aspect (through climate and vegetation) have an influence upon the spatial distribution and density of the sand fly fauna[11,12].

In Morocco, other ecological factors are studied in their relationship with sand fly abundance and distribution such urbanization, proximity of humans and domestic animals, organic matter in the soil, shelter and vegetation type[13,14]. In southern Morocco, little is known about species composition of sand fly fauna. Epidemiological situation of leishmaniasis, especially in extreme south, needs to be updated. Actually, zoonotic cutaneous leishmaniasis (ZCL) has spread endemically in the southern Ouarzazate region and in the northern oriental highlands. This form is caused by *L. major*, and transmitted by *P. papatasi*, with *Meriones shawi* as the main reservoir host[15].

Rioux et al. have led an entomological survey carried out in the littoral zone of the western Saharan region until Dakhla province[7]. Recently, entomological investigations were carried out on both the northern and southern slopes of the mountains until Ouarzazate province[16].

In the present work, we discuss the possible effect of many ecological factors on the diversity and distribution of Moroccan sandflies through three transects: Ouarzazate-M’Hamid, Fom Zguid-Marrakesh and Erfoud-Nador (Table 1 and Figure 1). We give a particular attention to ZCL foci (Ouarzazate, Zagora, Tata and Errachidia) with the aim to update their entomological data.

Table 1

Information about the study areas showing the 29 sandfly collection sites.

Site (code)	Co-ordinates	Elevation above sea level (m)
Ouarzazate (S1)	30°50'N 6°46'W	1 100
Tisserghate (S2)	30°40'N 6°21'W	920
Tinzoulina (S3)	30°39'N 6°20'W	803
Beni zoli (S4)	30°25'N 5°91'W	760
Zagora (S5)	30°21'N 5°48'W	732
Tamegroute (S6)	30°13'N 5°39'W	676
Zaouia Sidi Moktar (S7)	30°11'N 5°34'W	371
Anagam (S8)	30°13'N 5°41'W	653
Tagounite (S9)	30°20'N 5°50'W	642
M’Hamid (S10)	29°49'N 5°43'W	552
Fom Zguid (S11)	30°4'N 6°52'W	645
Tata (S12)	29°44'N 7°58'W	693
Issafn (S13)	29°51'N 8°31'W	1 141
Igherm (S14)	30°5'N 8°27'W	1 720
Ait aiaza (S15)	30°30'N 8° 47'W	282
Ouled Berhil (S16)	30°38'N 8°28'W	485
Tinmel (S17)	30°59'N 8°30'W	1 247
Ouirgane (S18)	31°10'N 8°4'W	931
Tahanaout (S19)	31°21'N 7°57'W	925
Marrakech (S20)	31°35'N 8°0'W	482
Erfoud (S21)	31°26'N 4°14'W	802
Errachidia (S22)	31°55'N 4°25'W	1 030
Midelt (S23)	32°40'N 4°43'W	1 470
Azrou (S24)	33°5'N 5°13'W	1 250
El Hajeb (S25)	33°41'N 5°22'W	985
My Yaacoub (S26)	34°5'N 5°10'W	260
Chefchaouen (S27)	35°10'N 5°16'W	593
Al Hoceima (S28)	35°14'N 3°56'W	76
Nador (S29)	35°10'N 2°55'W	61

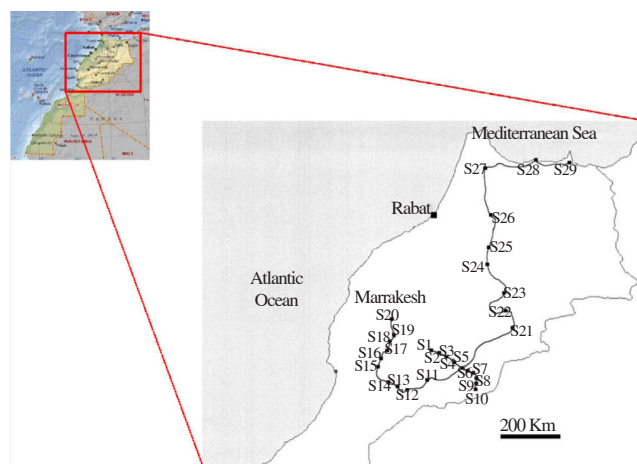


Figure 1. Study area showing the 29 sandfly collection sites.

2. Materials and methods

2.1. Transects

Three transects were studied. On June 2010, the first transect was 267 km long and linked the cities of Ouarzazate and M’Hamid. This transect went through eight collection sites, where altitude varies from 552 to 1 100 m, on the southern slopes of the High Atlas Mountains. The second transect was 537 km long from Fom Zguid to Marrakesh city and was studied on June 2011. This transect went through eight collection sites where altitude varies between 282 and 1 720 m. The third and last transect was 1 006 km

long and connected Erfoud city to the city of Nador. This transect went through seven collection sites with altitude ranges from 61 to 1470 m and was studied on June 2012 (Figure 1).

2.2. Sandfly collection and identification

Sandflies were collected using sticky paper traps (each an A4 sheet of paper coated with castor oil) for one night. Specimens were preserved in 70% ethanol, cleared in 20% (w/v) KOH and Marc-André solution, and then mounted on microscope slides, in Hoyer's medium.

Most of the sandflies were then identified to species by the morphological examination of the genitalia (males) or the pharyngeal armature and spermathecae (females), with the help of the keys and descriptions published by Moroccan Ministry of Health[3,17].

For *P. perniciosus* complex, the females were identified by examining the dilatation of distal parts of spermathecal ducts and males by examining the shape of the copulatory valves and counting the number of coxite hairs[18,19].

2.3. Data analysis

Various ecological parameters were calculated to characterize the sand fly populations in the different sites and habitats:

$$\text{Relative frequency} = \frac{\text{Number of specimens of one species}}{\text{Total of specimens}} \times 100\%$$

$$\text{Biodiversity index} = (S-1)/\log N$$

Where N is the number of individuals, S refers to the number of species in the sample.

Sorensen index (Sorensen's similarity coefficient) which can have values from 0 (no similarity) to 1.0 (complete similarity) = $2a/2a+b+c$ (a: number of species in both sample A and sample B, b: number of species in sample A but not in sample B and c: number of species in sample B but not in sample A).

3. Results

Overall, 7140 sandflies were collected along the three transects. In the combined collections, nine *Phlebotomus* species: *P. papatasi* (27.6%), *P. longicuspis* (19%), *P. sergenti* (18.2%), *P. perniciosus* (6.2%), *Phlebotomus bergeroti* (*P. bergeroti*) (2.9%), *Phlebotomus alexandri* (*P. alexandri*) (1.4%), *Phlebotomus chadlii* (*P. chadlii*) (0.8%), *Phlebotomus chabaudi* (*P. chabaudi*) (0.5%) and *Phlebotomus ariasi* (*P. ariasi*) (0.5%) and five *Sergentomyia* species: *Sergentomyia minuta* (*S. minuta*) (10.4%), *Sergentomyia fallax* (*S. fallax*) (8.1%), *Sergentomyia dreyfussi* (*S. dreyfussi*) (2.1%), *Sergentomyia christophersi* (*S. christophersi*) (1.7%) and *Sergentomyia africana* (*S. africana*) (0.5%) were detected. The detail of sampling in each transect is shown respectively in Tables 2-4.

Among the 2056 sandflies collected on the Ouarzazat-M'Hamid road, *P. longicuspis* was the most common species (25.5%), followed by *P. papatasi* (23.8%), *S. fallax* (12.6%), *P. sergenti* (16.6%), *S. minuta* (11%), *P. bergeroti* (5.4%), *S. dreyfussi* (1.8%), *S. christophersi* (1.7%) and *P. ariasi*, *P. perniciosus*, *P. alexandri*, *P. chabaudi* and *P. chadlii* with less than 1% each. The sex ratio was in favor of males for all species in all stations with the exception of *S. minuta* (M/F=13/14) in Tisserghate station and *P. chadlii* (0/1) in Ouarzazat city (Table 2).

On the Fom Zguid-Marrakesh road, *P. papatasi* was the most prevalent species (25.2%) of a total of 2370 sandflies collected, followed by *P. longicuspis* (17.5%), *S. minuta* (16.2%), *P. perniciosus* (11.7%), *P. sergenti* (11.7%), *S. fallax* (7.1%), *S. dreyfussi* (3.1%), *S. christophersi* (2.2%), *P. bergeroti* (1.9%) and *P. alexandri* (1.1%). We collected *S. africana*, *P. chabaudi*, *P. ariasi* and *P. chadlii* with less than 1% each as well. The sex ratio was in favor of females for *P. ariasi* in Ait aiaaza, *S. dreyfussi* and *S. christophersi* in Tata city, while, it was in favor of males for all caught species in all other stations (Table 3).

Sandflies (2714) were collected on the Erfoud-Nador road.

Table 2

Number of males (M) and females (F) sandflies collected along the Ouarzazate-M'Hamid transect.

Species	<i>P. papatasi</i>		<i>P. bergeroti</i>		<i>P. sergenti</i>		<i>P. alexandri</i>		<i>P. chabaudi</i>		<i>P. ariasi</i>		<i>P. perniciosus</i>		<i>P. longicuspis</i>		<i>P. chadlii</i>		<i>S. fallax</i>		<i>S. minuta</i>		<i>S. dreyfussi</i>		<i>S. christophersi</i>		Total
Site (code)	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	
Ouarzazate (S1)	36	15	0	0	24	11	3	1	2	1	11	2	0	0	43	24	0	1	12	3	12	2	1	3	0	0	207
Tisserghate (S2)	106	13	0	0	144	25	0	0	0	0	2	0	5	1	224	20	0	0	18	15	13	14	0	0	0	0	600
Tinzouline (S3)	52	11	1	0	20	2	0	0	0	0	0	0	2	0	123	6	0	0	19	7	12	10	0	0	2	0	267
Beni zoli (S4)	30	12	14	5	13	11	0	0	0	0	0	0	0	0	10	2	0	0	23	8	19	3	0	0	3	1	154
Zagora (S5)	23	7	15	3	20	9	0	0	0	0	0	0	0	0	2	0	0	0	9	2	11	9	2	1	6	1	120
Tamegroute (S6)	28	10	30	10	6	0	0	0	0	0	0	0	0	0	12	2	0	0	12	4	21	2	0	0	9	0	146
Zaouia Sidi Moktar (S7)	33	18	11	1	4	2	0	0	0	0	0	0	0	0	3	0	0	0	15	7	16	7	7	0	5	3	132
Anagam (S8)	22	4	8	0	10	2	1	0	0	0	0	0	0	0	20	8	0	0	18	11	23	11	2	0	2	0	142
Tagounite (S9)	19	8	10	2	16	3	0	0	0	0	0	0	0	0	16	4	0	0	23	2	18	9	10	4	1	0	145
M'Hamid (S10)	30	12	1	0	15	4	0	0	0	0	0	0	0	0	5	0	0	0	33	18	12	2	7	1	3	0	143
Total	M/F																										
All	489		111		341		5		3		15		8		524		1		259		226		38		36		2056

Table 3

Number of males (M) and females (F) sandflies collected along the Fom Zguid-Marrakech transect.

Species	<i>P. papatasi</i>		<i>P. bergeroti</i>		<i>P. sergenti</i>		<i>P. alexandri</i>		<i>P. chabaudi</i>		<i>P. ariasi</i>		<i>P. perniciosus</i>		<i>P. longicuspis</i>		<i>P. chadlii</i>		<i>S. fallax</i>		<i>S. minuta</i>		<i>S. dreyfussia</i>		<i>S. africana</i>		<i>S. christophersi</i>		Total	
Site (code)	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F		
Foum Zguid	60	18	18	3	24	8	4	1	0	0	0	0	0	0	4	0	0	0	10	2	24	13	0	0	4	1	10	2	180	
(S11)																														
Tata (S12)	88	61	9	0	36	14	12	2	0	0	0	0	0	0	2	0	0	0	8	5	38	21	0	1	12	4	4	6	323	
Issafn (S13)	24	16	3	0	23	16	3	1	11	5	3	1	14	3	16	8	4	2	0	0	17	5	13	4	0	0	0	0	182	
Igherm (S14)	0	0	0	0	2	0	0	0	0	0	3	1	8	1	24	18	0	0	2	0	3	1	0	0	0	0	0	0	63	
Ait aiaaza	18	6	4	0	4	1	2	0	0	0	0	1	0	0	2	0	0	0	13	3	27	5	3	1	0	0	18	11	119	
(S15)																														
Ouled Berhil	32	14	6	2	22	8	0	0	0	0	0	0	0	0	6	0	0	0	43	19	83	49	14	6	0	0	0	0	304	
(S16)																														
Tinmel (S17)	25	11	0	0	13	6	2	0	0	0	2	0	14	6	33	12	0	0	0	0	8	0	0	0	0	0	0	0	132	
Ouirgane	16	10	0	0	8	2	0	0	0	0	0	0	33	18	43	11	0	0	7	2	20	12	5	0	0	0	0	0	187	
(S18)																														
Tahanaout	23	14	0	0	10	7	0	0	0	0	0	0	115	65	123	77	0	0	11	9	16	2	3	1	0	0	0	0	466	
(S19)																														
Marrakech	90	72	0	0	50	23	0	0	0	0	0	0	0	0	22	13	0	0	24	11	28	12	17	6	0	0	0	0	348	
(S20)																														
Total	M/F	376/222		40/5		192/85		23/4		11/5		8/3		184/93		275/139		4/2		118/51		264/120		55/19		16/5		32/19		1558/746
All		598		45		277		27		16		11		277		414		6		169		384		74		21		51	2370	

Table 4

Number of males (M) and females (F) sandflies collected along the Erfoud-Nador transect.

Species	<i>P. papatasi</i>		<i>P. bergeroti</i>		<i>P. sergenti</i>		<i>P. alexandri</i>		<i>P. chabaudi</i>		<i>P. ariasi</i>		<i>P. perniciosus</i>			<i>P. longicuspis</i>		<i>P. chadlii</i>		<i>S. fallax</i>		<i>S. minuta</i>		<i>S. dreyfussi</i>		<i>S. Africana</i>		<i>S. christophersi</i>		Total	
Site (code)	M	F	M	F	M	F	M	F	M	F	M	F	M	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F		
Erfoud (S21)	22	13	14	6	1	0	4	1	0	0	0	0	(PN) (PNA)		0	0	0	0	0	0	17	11	0	0	3	1	7	1	12	4	117
Errachidia (S22)	56	34	8	2	12	8	40	17	2	1	0	0	0	0	0	4	1	1	0	11	4	5	2	0	2	4	0	0	0	214	
Midelt (S23)	23	12	3	0	19	3	0	0	6	2	2	0	19	8	14	31	26	13	2	11	3	5	1	0	1	0	0	0	0	204	
Azrou (S24)	88	53	0	0	76	42	3	1	4	1	6	2	12	5	10	56	29	11	6	2	0	11	7	1	1	0	0	0	0	427	
El Hajeb (S25)	72	39	0	0	114	68	2	0	2	0	0	0	39	17	23	53	18	7	1	7	3	12	5	12	9	0	0	0	0	503	
My Yaacoub (S26)	263	138	0	0	172	91	0	0	0	0	0	0	8	4	6	101	83	2	0	28	15	22	5	7	2	0	0	0	0	947	
Chefchaouen (S27)	45	33	0	0	27	18	0	0	1	1	0	0	4	1	0	16	3	4	0	12	3	16	9	0	1	0	0	0	0	194	
Al Houceima (S28)	12	3	0	0	16	4	0	0	0	0	0	0	0	0	0	0	0	0	13	2	4	0	0	0	0	0	0	0	0	54	
Nador (S29)	11	6	0	0	12	8	0	0	0	0	0	0	0	0	0	0	0	0	2	0	12	3	0	0	0	0	0	0	0	54	
Total	M/F	592/331		15/8		449/242		49/19		15/5		8/2		117/53		261/160		38/9		103/41		87/32		23/17		11/1		12/4		1790/924	
All		923		33		691		68		20		10		170		421		47		144		119		40		12		16	2714		

P. papatasi was the most prevalent species (34%), followed by *P. sergenti* (25.5%), *P. longicuspis* (15.5%), *P. perniciosus* (6.3%), *S. fallax* (5.3%), *S. minuta* (4.4%), *P. alexandri* (2.5%), *P. chadlii* (1.7%), *P. bergeroti* (1.2%) and *S. dreyfussii* (1.5%). *S. africana*, *S. christophersi*, *P. chabaudi* and *P. ariasi* with less than 1% each. Except for *S. dreyfussii* (in Errachidia, Midelt, Azrou and Chefchaouen) for each sandfly species, the males collected outnumbered the females in all stations (Table 4).

In the first and second transects, *P. perniciosus* males were collected only as atypical morphology (PNA), but in Erfoud-Nador road, both forms *P. perniciosus* typical (PN) and *P. perniciosus* atypical were collected in the same stations (Tables 2-4).

For ZCL foci, a total of 207 specimens were collected in Ouarzazat, 120 in Zagora, 323 in Tata and 214 in Errachidia city. *P. longicuspis* was the most abundant species in Ouarzazat with 32% while *P. papatasi* was the most prevalent once in Zagora, Tata and Errachidia with 25%, 46% and 42% respectively. Each of the 29 study sites was highly similar to the other sites with Sorensen similarity indexes varying from 0.74 to 1 (complete similarity).

Table 5 summarizes the numbers of sand fly species and individuals caught in each category of biotope and the

corresponding sand fly biodiversity indices. If it is not null, the sand fly fauna biodiversity of peridomestic biotope of Issafn (S13) showed the greatest value while that in the Barbacanes of Al Houceima (S28) showed the least one.

Table 5

Sandflies caught in different biotopes along the 29 sites studied with the corresponding biodiversity index.

Code	Biotope	S	Individuals	Biodiversity index	Dominant species
S1	Ruin	4	27	0.91	<i>P. longicuspis</i>
	Peridomestic	10	170	1.75	<i>P. longicuspis</i>
S2	Peridomestic	7	307	1.05	<i>P. longicuspis</i>
	Intradomestic	7	280	1.06	<i>P. papatasi</i>
S3	Barbacanes	4	19	1.02	<i>P. longicuspis</i>
	Barbacanes	5	92	0.88	<i>P. papatasi</i>
S4	Peridomestic	8	175	1.36	<i>P. longicuspis</i>
	Intradomestic	6	30	1.47	<i>P. papatasi</i>
S5	Vegetation	5	44	1.06	<i>P. papatasi</i>
	Stable	6	80	1.14	<i>P. papatasi</i>
S6	Stable	7	66	1.43	<i>P. papatasi</i>
	Intradomestic	6	18	1.73	<i>P. papatasi</i>
S7	Barbacanes	4	16	1.08	<i>P. papatasi</i>
	Peridomestic	7	103	1.29	<i>P. papatasi</i>
S8	Ruin	6	43	1.33	<i>P. bergeroti</i>
	Peridomestic	7	83	1.36	<i>P. papatasi</i>
S9	Vegetation	5	39	1.09	<i>S. fallax</i>
	Wall	4	29	0.89	<i>P. papatasi</i>
S10	Peridomestic	8	81	1.59	<i>P. longicuspis</i>
	Rock crevices	5	32	1.15	<i>S. minuta</i>

Table 5, continued

Sandflies caught in different biotopes along the 29 sites studied with the corresponding biodiversity index.

Code	Biotope	S	Individuals	Biodiversity index	Dominant species
S9	Peridomestic	7	73	1.40	<i>S. fallax</i>
	Stable	6	45	1.31	<i>P. papatasi</i>
	Rock crevices	4	27	0.91	<i>S. minuta</i>
S10	Peridomestic	8	143	1.41	<i>S. fallax</i>
S11	peridomestic	8	102	1.51	<i>P. papatasi</i>
	Barbacanes	5	45	1.05	<i>P. sergenti</i>
	Wall	5	33	1.14	<i>S. minuta</i>
S12	Peridomestic	8	223	1.29	<i>P. sergenti</i>
	Barbacanes	7	76	1.39	<i>P. papatasi</i>
	Wall	6	24	1.57	<i>S. minuta</i>
S13	Peridomestic	11	182	1.92	<i>P. sergenti</i>
S14	Peridomestic	6	40	1.36	<i>P. longicuspis</i>
	Intradomestic	3	23	0.64	<i>P. longicuspis</i>
S15	Intradomestic	6	79	1.14	<i>P. papatasi</i>
	Peridomestic	8	40	1.90	<i>S. minuta</i>
	Intradomestic	7	183	1.15	<i>P. papatasi</i>
S16	Vegetation	6	40	1.36	<i>S. fallax</i>
	Barbacanes	6	81	1.14	<i>S. minuta</i>
S17	Peridomestic	7	52	1.52	<i>P. longicuspis</i>
	Wall	6	33	1.43	<i>P. longicuspis</i>
	Barbacanes	7	47	1.56	<i>P. longicuspis</i>
S18	Peridomestic	7	93	1.32	<i>P. papatasi</i>
	Barbacanes	6	74	1.16	<i>P. longicuspis</i>
	rock crevices	4	20	1.00	<i>P. longicuspis</i>
S19	Intradomestic	5	277	0.71	<i>P. perniciosus</i>
	Peridomestic	7	189	1.14	<i>P. longicuspis</i>
S20	Barbacanes	5	217	0.74	<i>P. papatasi</i>
	Peridomestic	5	101	0.87	<i>P. papatasi</i>
	Vegetation	4	30	0.88	<i>P. papatasi</i>
S21	Intradomestic	6	593	0.78	<i>P. papatasi</i>
	Peridomestic	8	231	1.29	<i>P. bergeroti</i>
	Vegetation	5	95	0.88	<i>S. fallax</i>
S22	Intadomestic	6	754	0.75	<i>P. papatasi</i>
	Barbacanes	8	179	1.35	<i>P. alexandri</i>
	Peridomestic	10	311	1.57	<i>P. papatasi</i>
S23	Peridomestic	11	1374	1.38	<i>P. perniciosus</i>
	Intradomestic	6	300	0.88	<i>P. longicuspis</i>
S24	Peridomestic	11	976	1.45	<i>P. longicuspis</i>
	Intradomestic	7	701	0.92	<i>P. perniciosus</i>
S25	intradomestic	6	373	0.84	<i>P. longicuspis</i>
	Vegetation	5	130	0.82	<i>P. sergenti</i>
	Peridomestic	10	985	1.31	<i>P. papatasi</i>
S26	Peridomestic	8	711	1.07	<i>P. longicuspis</i>
	Intradomestic	6	320	0.87	<i>P. sergenti</i>
	Barbacanes	8	176	1.35	<i>P. papatasi</i>
S27	Peridomestic	9	377	1.35	<i>P. sergenti</i>
	Intradomestic	5	261	0.72	<i>P. papatasi</i>
	Wall	6	149	1.00	<i>S. minuta</i>
S28	Peridomestic	4	76	0.69	<i>P. papatasi</i>
	Intradomestic	0	0	0.00	0
	Barbacanes	3	54	0.50	<i>P. sergenti</i>
S29	Intradomestic	0	0	0.00	0
	Peridomestic	4	74	0.70	<i>P. sergenti</i>

Some qualitative and quantitative differences between biotopes were noted. In terms of number and specific richness of sand fly fauna, we collected 11 species in peridomestic biotope of S13, S23 and S24 whereas only three species in intradomestic biotope and barbacanes of S14 and S28 respectively.

We noted also that most of the *Sergentomyia* species collected were from sites far from human residences like wall, rock crevices, vegetations and peridomestic habitats.

4. Discussion

In Morocco, twenty three species of phlebotomine sandflies have been reported; 14 species of *Phlebotomus* genus and 9 species of *Sergentomyia* genus[2] of which five species have been known to transmit the disease: *P. ariasi*, *P. perniciosus* and *P. longicuspis* are vectors of *L. infantum* and are spread mainly in northern regions[19], *P. papatasi* is vector of humid form of cutaneous leishmaniasis caused by *L. major* seen in the south and southeast of the Atlas Mountains, while *P. sergenti* is the vector for the dry skin form of cutaneous leishmaniasis caused by *L. tropica*, reported in the center of the country[20-22].

In this study, 14 sandfly species were identified, representing 60% of the Moroccan sand fly species. Two genera are identified; *Phlebotomus* (77.2%) and *Sergentomyia* (22.8%) and all vectors are well represented.

The five vectors - *P. papatasi*, *P. ariasi*, *P. perniciosus*, *P. longicuspis* and *P. sergenti* - made up 71% of the sandflies collected, while in 86% of study sites (25/29), three of them - *P. papatasi*, *P. sergenti* and *P. longicuspis* - coexist.

We suggest that leishmaniasis in Morocco is more related to the parasite ecology rather than the vector distribution, even if the vector occurrence is very important. Role of different vectors are often determined by species-level co-evolution of susceptibility to *Leishmania* species, with selection being initiated and maintained by ecological contacts[23].

Vector-borne and zoonotic disease transmission risk is the result of interactions between different species in space and time[24]. But, many entomological investigations in Morocco show no correlation between these vectors distribution and disease distribution even if the hosts and reservoir playing a role in the transmission of *Leishmania* are present in both endemic and non-endemic area[11,24,25].

The distribution of *P. papatasi* extends from southern Europe and eastern regions to the Indian subcontinent and highly depends on environment factors particularly relative humidity and temperature. It thus largely exceeds those of *L. major*[11]. In Egypt for example, ZCL is primarily present in northern Sinai while the vector distribution is more extensive[26]. Until now, despite the wide distribution of both *P. papatasi* and *Meriones shawi* in Morocco, ZCL and *L. major* are restricted to the pre-Saharan areas.

In Iran, Yavar *et al.* (2013) found natural infection of *P. papatasi* by *L. infantum*, with the absence of human VL cases. This vector has the ability to transmit two species of *Leishmania* parasite: *L. major* and *L. infantum*[27].

P. sergenti has also an extensive geographical distribution, wider than that of the anthroponotic cutaneous leishmaniasis[28,29]. In Portugal, *P. sergenti* showed a very short period of activity in

comparison with other sandflies[30].

It is suggested that the presence of *P. sergenti* in *L. tropica* free areas is related to the existence of some cryptic vector species with consequences in their capacity to readily transmit *Leishmania*[31] and/or mechanisms of transmission for *Leishmania* parasites[32]. In Spain, two *P. sergenti* lineages were identified, a typically Spanish mitochondrial lineage and another one that is common in Morocco[28,29].

Phlebotomus ariasi, *P. perniciosus* and *P. longicuspis* are considered as *L. infantum* vectors[4]. The vectorial role of *P. ariasi* has been described in northern Morocco[33] when, previous studies[18,19,34,35] showed the presence in Morocco of three phylogenetic species of *P. perniciosus* complex: *P. perniciosus* including PN and PNA-like morphs; *P. longicuspis* sensu stricto and a sibling species of *P. longicuspis* (LCx). The atypical morph of *P. perniciosus* was identified also in Tunisia[36] and in Algeria[37]. The vectorial capacity and competence of these *P. perniciosus* morphs as well as of *P. longicuspis* and its potential cryptic species should be confirmed.

L. infantum is widespread mainly in northern Morocco as causative agent of cutaneous and visceral leishmaniasis with sporadic cases of human visceral leishmaniasis in the south[20,38]. Boussaa *et al.* (2008) noted the same distribution of *P. perniciosus* forms[19]. In North of Morocco, typical morphs of *P. perniciosus* are the most abundant forms, while in the southern regions, are mainly an atypical form.

Our results confirm this distribution, atypical morphology of *P. perniciosus* males was collected in the first and second transects, but in Erfoud-Nador road, both forms were collected in the same stations. Monitoring the insects in natural ecosystem is one of the simplest ways to observe and provide early warning of changes to biodiversity and habitat structure[39].

In our study, species presence and diversity appeared to be affected by several factors. According to altitude, which acting through its relationship with climate, we noted relatively low biodiversity at the highest (1 720 m) and low-altitude (61 m) sites investigated. Jahanifard *et al.* (2014) did find similar results in Iran[40], showing *P. papatasi* to be found between 8 and 1 756 m[41-43]. In contrast, some species prefer high altitude such as *P. ariasi* collected only between 920 and 1 720 m. Other species change their sites according to bioclimate. For example, *P. chabaudi* and *P. chadlii* were collected at high altitude in arid climate and low altitude with humid climate.

On the other hand, a high diversity level in Saharan areas has been found when the species are structured on bioclimate basis[44]. Our results are similar to the other authors that found a high diversity of sandflies in sites ranged from 900 to 1 100 m[11,16,45]. Nador and

Al Hoceima are two coastal cities and show a low specific richness level.

In conclusion, the wide distribution of vectors in different forms of leishmaniasis in Morocco could increase the risk to spread of different species of *Leishmania* from many foci to nonendemic sites. This suggests the need for a continuously surveillance to control the situation in these foci and to prevent risk in nonendemic areas.

Conflict of interest statement

We declare that we have no conflict of interest.

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Comments

Background

Human CL is a serious public-health problem in Morocco with active foci from 1970s. Several faunistic studies related to the distribution of phlebotomine sandflies in these foci have provided important data by describing the species involved in CL. The study on populations of sandflies by entomological surveys and the climatic factors that could affect their distribution need to be studied in new foci of CL.

Research frontiers

This type of works contributes to the need for appropriate control programs for Leishmaniasis in endemic countries.

Related reports

There are many reports related to this research in other regions of Morocco. In overall, in the three transects studied, nine *Phlebotomus* species were found. *P. papatasi* was the most prevalent species responsible of *L. major* spreading and human CL.

Innovations & breakthroughs

This is the first contribution in which entomological studies are related with ecological factors from Northern to Southern of Morocco where twenty-nine sampling stations were studied along three transects with a high diversity of *Phlebotomus* species found.

Applications

Entomological surveys in areas where leishmaniasis foci occur are imperative for the control of the disease. This work contributes to a better understanding of the distribution of the different species of sandflies present along three transects in Morocco in relation to ecological factors to undertake better control programs in this country.

Peer review

In this work authors conducted an entomological survey along three transects of Morocco where foci of cutaneous leishmaniasis occur. The data obtained gives practical information and suggests the need for a continuous entomological surveillance in this country.

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